Review: Research Design & Advanced Topics

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Applied Quantitative Methods II

MA in Social Sciences, Spring 2026

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  - → Causal inference methods beyond FE/DiD
  - → Defending your results: robustness and validity
  - → Common pitfalls when applying all of this
- Plus a pointer to two advanced topics (not covered today)

Roadmap

From estimation to identification

Causal inference toolkit

Defending a causal claim

Choosing the right method & pitfalls

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- Plus computing practices to make all of this reproducible

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Identification: do those parameters mean what I think they mean?

Your R output answers the first.

Only your research design can answer the second.

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- Causal inference = principled way to estimate the missing counterfactual

What we can estimate on average

- We can't recover τ_i for individuals
- But we can target averages:
  - → **ATE —** average over everyone: $E[Y(1) - Y(0)]$
  - → **ATT —** average over the treated: $E[Y(1) - Y(0) \mid D = 1]$
  - → **LATE —** average over those who *respond* to a treatment (IV, RDD)
- Different methods estimate different quantities
  - → When you compare two papers: are they even targeting the same thing?

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  - $\rightarrow$ **Collider** $D \rightarrow C \leftarrow Y$
- What you control for — and what you do **not** — depends on which structure you are looking at

Confounding: the standard case

$$\begin{tikzcd} & \text{Confounder } Z \arrow[dl] \arrow[dr] & \\ \text{Treatment } D \arrow[rr] & & \text{Outcome } Y \end{tikzcd}$$

- Z affects both D and Y: a back-door path
- Un-adjusted comparison of treated vs. untreated mixes two effects
- **Solution:** condition on Z (or use a design that blocks the back door)

Colliders: do not control!

$$
\begin{tikzcd}
& \text{Collider } C & \\
\text{Treatment } D \arrow[ur] \arrow[rr] & & \text{Outcome } Y \arrow[ul]
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$$

- C is caused *by* both D and Y
- Conditioning on C creates a spurious association between D and Y
- Classic: "smart people lack social skills" (if you condition on being hired)

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- Rule of thumb: if in doubt about whether a variable is pre- or post-treatment, leave it out

So what does your regression assume?

Every regression coefficient you report implicitly assumes:

1. You have **closed all back doors** (no omitted confounders)
2. You have **not opened new ones** (no controlling for colliders or mediators)
3. The functional form is roughly right (linearity, additivity)

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(1) and (2) are **design** assumptions, not something you can test.

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What you already have

Two of the four big causal-inference designs are already in your toolkit:

- **Fixed effects** — control for all time-invariant unobservables at some level (unit, region, year. . .)
- **Difference-in-differences** — use a never-treated comparison group plus pre/post variation; identifies the ATT under parallel trends

Two more that you should at least recognize:

- **Regression discontinuity design (RDD)**
- **Instrumental variables (IV)**

Regression discontinuity design

- **Setup:** treatment assignment depends on whether a *running variable* crosses a cutoff

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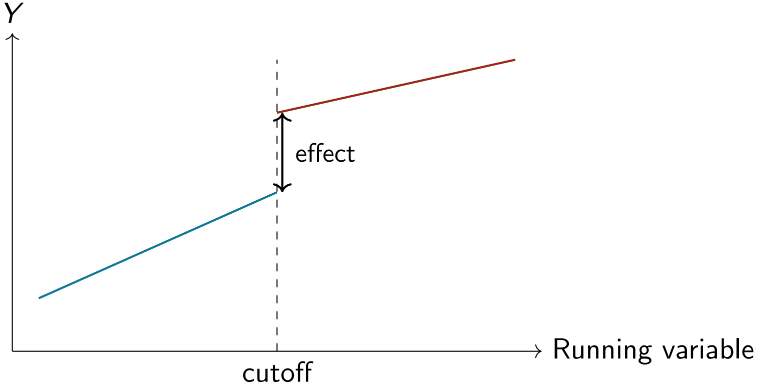
- Estimates a LATE: effect for units near the threshold

RDD visually



- The jump at the cutoff is the treatment effect
- The trend on each side controls for whatever else varies smoothly

Instrumental variables

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  - → **Relevance**: Z actually affects D (testable)
  - → **Exclusion restriction**: Z affects Y only through D (not testable)

The four design templates

Design	Exogenous variation	Key assumption
FE	Within-unit differences	No time-varying confounders
DiD	Treated vs. comparison, pre/post	Parallel trends
RDD	Jump at cutoff	No manipulation at cutoff
IV	Instrument $\rightarrow$ treatment	Exclusion restriction

- You do not pick a design, *the data* pick one for you
- What source of exogenous variation does the world hand you?

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- Often used *with* regression or DiD, not instead of

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Your main result is the beginning

- You have a point estimate, a standard error, a figure
- A skeptical reader should be asking:
  - → What if you had specified it differently?
  - → What if your identifying assumption is wrong?
  - → What if you are measuring the wrong thing?
  - → Would the result generalize elsewhere?
- Your job is to answer those questions **before they do**

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- Display as a coefficient plot across specifications, not a 10-column table no one reads

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  - → Post-period coefficients show the dynamic effect

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- A paper that only shows the main effect is weaker than one that shows the main effect **plus** two or three such auxiliary findings

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- Identification can be perfect and the claim still be wrong if the measurement doesn't line up with the theory

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- **Temporal validity** (Munger 2023): does it still hold 10 years from now?

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Starting point: your question type

Goal	Tools
Describe a pattern	Summary stats, visualization, simple regression
Predict Y from X	Regression, ML (cross-validated)
Explain Y causally	Design-based methods: FE, DiD, RDD, IV, experiments
Generalize a mechanism	Theory + replication across contexts

- Most confusion in applied work comes from mixing these up
  - $\rightarrow$ e.g. treating predictive coefficients as causal

Starting point: your outcome

Outcome type	**Method**
Continuous	OLS (always a good first pass)
Binary	Logit, probit (or LPM for interpretability)
Count (non-negative integer)	Poisson, negative binomial
Ordered categories	Ordered logit / probit
Time-to-event	Cox, Kaplan–Meier, parametric survival
Panel data	+ fixed effects, + clustered SEs
Spatial data	+ spatial weights, + Moran's I

- This is a cheat-sheet for your final essay — keep it

Pitfall 1: p-hacking and forking paths

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- Mitigations: pre-analysis plans, specification curves, honest reporting of all tried specifications

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- Cinelli, Forney & Pearl (2024) "A crash course in good and bad controls"

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- Consequence: default SEs can be *much* too small
- Fixes: cluster-robust SEs, Conley SEs (spatial), network-adjusted SEs

What's next (not in this class)

- **Multilevel / hierarchical models** (Gelman & Hill)
  - → Natural framework when data are nested in groups
- **Bayesian inference**
  - → Priors, posteriors, and what MCMC actually does
  - → McElreath, *Statistical Rethinking*
- **Machine learning for social science**
  - → Prediction-first tools: lasso, random forests, boosting
  - → Causal ML (honest trees, double/debiased ML)
- **Text as data**
  - → From bag-of-words to LLMs in social science

A checklist for your final essay

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- How do you **measure** your key concepts, and what's lost in that translation?

Good luck with the essay.

Appendix follows: multilevel models and text as data.

Self-study material, not covered in class.

Roadmap

Appendix: Multilevel models

Appendix: Text as data

Hierarchical data

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  - → Groups vary in how they respond to predictors
- Ordinary OLS ignores both

Three ways to handle groups

- **Complete pooling**: ignore groups entirely, one intercept
  - $\rightarrow$ Wastes the group-level information
- **No pooling**: separate intercept for each group (fixed effects)
  - $\rightarrow$ Fits each group exactly — but noisily for small groups
- **Partial pooling**: each group gets its own intercept, but shrunk toward a global mean
  - $\rightarrow$ Best of both worlds — this is what multilevel models do

Random intercepts

$$y_{ij} = \alpha_j + \beta x_{ij} + \varepsilon_{ij}, \quad \alpha_j \sim \mathcal{N}(\mu_\alpha, \sigma_\alpha^2)$$

- Each group j has its own intercept α_j
- But the α_j's are drawn from a common distribution
- Estimate: one global mean + one variance, not 200 separate intercepts
- Versus classic FE: FE *estimates* each α_j; random intercepts *model* them

Random slopes

$$y_{ij} = \alpha_j + \beta_j x_{ij} + \varepsilon_{ij}$$

- Now *both* intercept and slope vary by group
- Effect of x differs across schools / countries / firms
- Heterogeneity is explicitly modeled, not averaged away
- Useful when you expect the effect itself to depend on context

Fitting multilevel models in R

- Frequentist / REML: `lme4`
  - → `lmer(y ~ x + (1 | group), data = d)` — random intercept
  - → `lmer(y ~ x + (1 + x | group), data = d)` — random slope
- Bayesian: `brms` or `rstanarm`
  - → Same formula syntax, full posterior draws
- Visualization: `sjPlot::plot_model()`, `broom.mixed`

Multilevel vs. fixed effects

	Multilevel	**Fixed effects**
Goal	Describe variation across groups	Remove group-level confounding
Assumes	Group effects uncorrelated with x	Nothing about that
Small groups	Handles well (shrinkage)	Noisy
Causal setting	Use if assumption plausible	Safer default

- If your goal is descriptive and you have many small groups, multilevel is natural
- If your goal is causal and you worry about group-level confounding, FE is safer

Multilevel: where to go next

- Gelman & Hill (2007) *Data Analysis Using Regression and Multilevel/Hierarchical Models*
- Gelman, Hill & Vehtari (2021) *Regression and Other Stories*, ch. 21–22
- McElreath *Statistical Rethinking*, ch. 13–14
- Bolker's `GLMM FAQ`: bbolker.github.io/mixedmodels-misc/glmmFAQ.html

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- A quantitative approach: turn documents into features, treat them like any other variable

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- Document these choices the way you would any other transformation

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- Classic in political science — see Laver, Benoit & Garry (2003) Wordscores; LIWC

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- R packages: `topicmodels`, `stm`

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  - → Replacing hand-coders for many tasks
  - → But: black box, bias, replication concerns

Text as data: where to go next

- Grimmer, Roberts & Stewart (2022) *Text as Data*. Princeton UP
  - → The standard reference for social scientists
- R packages worth knowing:
  - → `quanteda` — full preprocessing & analysis pipeline
  - → `stm` — structural topic models
  - → `text` — interface to transformer embeddings
- Course: Pablo Barberá's text-as-data materials (online)

Thanks, and good luck.